

Jacob Hailemariam

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EDUCATION

University of Calgary

Calgary, Alberta

Bachelor of Science in Electrical Engineering, Minor in Computer Engineering

Sep 2025 – May 2029

Cumulative GPA: 3.91/4.00 | First-Year GPA: 3.97/4.00

Relevant coursework: Programming with Data, Programming Fundamentals, Linear Methods, Digital Circuits

Awards: Schulich Dean's Entrance, President's Admission, Calgary Black Chambers Legacy Entrance

SKILLS

Languages: Python, C++, C, TypeScript, SQL

Machine Learning: PyTorch, NumPy, pandas, CNNs, Transformers, Mamba, Computer Vision, Multimodal Fusion

Technologies: Git, Linux, Docker, React, FastAPI, PostgreSQL, Redis, Zephyr RTOS, ARM Cortex-M, J-Link

EXPERIENCE

GSIL Machine Learning Researcher

Geospatial Sensing & Intelligence Lab | NSERC USRA Recipient

May 2026 – Present

- Design, build, and train deep learning models in **PyTorch** (CNNs, Transformers, Mamba) for **multimodal fusion** of **LiDAR** and **hyperspectral imagery**, writing custom dataset, transform, and training-loop code.
- Established a reproducible **benchmarking protocol** across three datasets on shared **Linux GPU servers** with **Git**; authoring a **first-author manuscript** for arXiv submission.
- Built a modular **Python** data pipeline that spatially aligns and transforms large remote-sensing datasets.

Avionics Software Engineer

Student Organization for Aerospace Research (SOAR)

Sep 2026 – Present

- Develop **C++** flight software for the Eos rocket avionics, interfacing **sensors** via **object-oriented** code.
- Contribute to **sensor drivers** and **telemetry** modules over **I2C**, **SPI**, and **UART** in a **Git**-based workflow.

Embedded in Embedded Member

Garmin-Affiliated Embedded Systems Program

Sep 2026 – Present

- Develop **embedded C** firmware on a **Nordic nRF52840** (ARM Cortex-M4) using **Zephyr RTOS**.
- Working through **GPIO**, **PWM**, **state machines**, **BLE**, and serial protocols with **J-Link** debugging.

PROJECTS

Vision Transformer for Image Classification

PyTorch, Python, Computer Vision, Deep Learning, Transformers

- Implemented a **13.4M-parameter Vision Transformer** from scratch with **multi-head self-attention**.
- Built the training pipeline with **MixUp/CutMix**, **RandAugment**, **EMA**, **mixed precision**, and LR warmup.
- Reached **89.63% test accuracy** on **CIFAR-10** without any pretrained weights.

Neural Network Chess Engine

C++17, PyTorch, React, Search Algorithms, Neural Networks

- Building a chess engine in **C++17** using **alpha-beta search** with a **neural network** position evaluator.
- Training the evaluator in **PyTorch** and integrating it into the **C++** search to score positions.
- Developing a **React** web app to play against the engine and view its evaluation of the position in real time.

LLM-Powered Study Planning Platform (EngInQuire)

TypeScript, React, Gemini API, LLMs, Prompt Engineering, Vercel

- Co-founded **EngInQuire** and built an **LLM** study planner on the **Gemini API** that generates weekly plans.
- Designed the **prompt pipeline** and a **structured-output parsing** layer that validates model responses.
- Shipped the platform in **TypeScript/React** on **Vercel** with a serverless booking flow, used by ~100 students.

LEADERSHIP

Junior Executive, External Affairs (Data Science & ML Club) | Junior Executive, Technology (CYBERSEC)

Junior Executive, Finance (EESA) | ColorStack Fellow | Peer Tutor | Youth Central Volunteer (150+ hours)